A SHARPNESS COUNTEREXAMPLE FOR THE λ_2 ENDPOINT IN A LIPSCHITZ PERTURBATION THEOREM

FA_BANACH_001, AGENT_LANE_12

Status

This packet answers Question 2.14 in the arXiv PDF of Krishna's paper [1]: the constant λ_2 in Theorem 2.13 cannot be improved beyond the endpoint 1. The result is a counterexample to any extension of the theorem that permits $\lambda_2 > 1$ while keeping the same invertibility conclusion.

Source question

The source theorem assumes, among other hypotheses, $\lambda_1 \in [0, 1)$ and $\lambda_2 \in [0, 1]$ in a Lipschitz perturbation inequality. Immediately after the proof, the source asks whether the constant λ_2 can be improved further. The crop below shows the question in the source PDF.

Date: 2026-06-20.

Then T is Lipschitz invertible. Further, for every $\varepsilon > 0$ satisfying $1 > \lambda_2 - \varepsilon > 0$ and $\lambda_1 + \varepsilon Lip(TS^{-1}) < 1$, we have

(i)
$$\frac{1-\lambda_1-\varepsilon Lip(TS^{-1})}{1+\lambda_2-\varepsilon}\|Sx-Sy\| \leq \|Tx-Ty\| \leq \frac{1+\lambda_1+\varepsilon Lip(TS^{-1})}{1-\lambda_2+\varepsilon}\|Sx-Sy\|, \quad \forall x,y \in \mathcal{X}.$$

(ii) *$\alpha S - T$ is invertible for all $\alpha \in \left(-\infty, \frac{1-\lambda_1-\varepsilon Lip(TS^{-1})}{1+\lambda_2-\varepsilon}\right)$.*

(iii) *T is invertible and*
$$\frac{1-\lambda_2+\varepsilon}{1+\lambda_1+\varepsilon Lip(TS^{-1})}\frac{1}{Lip(S)}\|u-v\| \leq \|T^{-1}u-T^{-1}v\| \leq \frac{1+\lambda_2-\varepsilon}{1-\lambda_1-\varepsilon Lip(TS^{-1})}Lip(S^{-1})\|u-v\|, \quad \forall u,v \in \mathcal{Y}.$$

(iv) *We have*
$$\frac{1-\lambda_1-\varepsilon Lip(TS^{-1})}{1+\lambda_2-\varepsilon}Lip(S) \leq Lip(T) \leq \frac{1+\lambda_1+\varepsilon Lip(TS^{-1})}{1-\lambda_2+\varepsilon}Lip(S) \quad \text{and}$$
$$\frac{1-\lambda_2+\varepsilon}{1+\lambda_1+\varepsilon Lip(TS^{-1})}\frac{1}{Lip(S)} \leq Lip(T^{-1}) \leq \frac{1+\lambda_2-\varepsilon}{1-\lambda_1-\varepsilon Lip(TS^{-1})}Lip(S^{-1}).$$

Proof. Define $R \coloneqq TS^{-1}$. Then for every $\varepsilon > 0$ satisfying $1 > \lambda_2 - \varepsilon > 0$ and $\lambda_1 + \varepsilon \mathrm{Lip}(TS^{-1}) < 1$,

$$\begin{aligned}
\|Ru-Rv-(u-v)\| &\leq \lambda_1\|u-v\| + \lambda_2\|Ru-Rv\| \\
&= \lambda_1\|u-v\| + (\lambda_2-\varepsilon)\|Ru-Rv\| + \varepsilon\|Ru-Rv\| \\
&\leq \lambda_1\|u-v\| + (\lambda_2-\varepsilon)\|Ru-Rv\| + \varepsilon\mathrm{Lip}(R)\|u-v\| \\
&= (\lambda_1+\varepsilon\mathrm{Lip}(R))\|u-v\| + (\lambda_2-\varepsilon)\|Ru-Rv\|, \quad \forall u,v \in \mathcal{Y}.
\end{aligned}$$

Remaining parts of the proof is similar to the proof of Theorem 2.7. □

It is an easy observation that the constant λ_1 in Theorem 2.13 can not be improved. We are therefore left with the following open problem.

Question 2.14. *Can the constant λ_2 be improved further in Theorem 2.13?*

3. Applications

Our first two applications of Theorem 2.9 are easy proofs of Soderlind-Campanato perturbation and Barbagallo-Ernst-Thera perturbation.

Theorem 3.1. *[6,45] (**Soderlind-Campanato perturbation**) Let $\mathcal{X}$ be a real Banach space, $A : \mathcal{X} \to \mathcal{X}$ be a map and there exist $\alpha > 0, 0 \leq \beta < 1$ such that*
$$\|\alpha Ax - \alpha Ay - (x-y)\| \leq \beta\|x-y\|, \quad \forall x,y \in \mathcal{X}.$$
Then A is Lipschitz, invertible and $Lip(A^{-1}) \leq \frac{\alpha}{1-\beta}$.

Proof. Set $T = \alpha A$ and $\lambda_1 = \beta$ in Theorem 2.9. Then $\frac{1}{\alpha}\mathrm{Lip}(A^{-1}) = \mathrm{Lip}(T^{-1}) \leq \frac{1}{1-\lambda_1} = \frac{1}{1-\beta}$. □

Theorem 3.2. *[2] (**Barbagallo-Ernst-Thera perturbation**) Let $\mathcal{X}$ be a real Banach space, $A : \mathcal{X} \to$*

Idea of the proof

The endpoint obstruction is the same geometric obstruction that separates a bi-Lipschitz embedding from a Lipschitz isomorphism onto the whole space. The unilateral shift on ℓ_2 is an isometric embedding with codimension-one range. Multiplying it by a large scalar makes the term $\lambda_2\|Tx-Ty\|$ dominate the perturbation inequality as soon as $\lambda_2 > 1$, but no scalar multiple of the unilateral shift is surjective. Thus the perturbation inequality cannot force invertibility past the endpoint $\lambda_2 = 1$.

Counterexample

Theorem 1. *Let $\lambda_2 > 1$ and let $0 \leq \lambda_1 < 1$. There are Banach spaces $\mathcal{X} = \mathcal{Y}$, an invertible Lipschitz map $S : \mathcal{X} \to \mathcal{Y}$, and a Lipschitz map $T : \mathcal{X} \to \mathcal{Y}$ such that*
$$\|Tx-Ty-(Sx-Sy)\| \leq \lambda_1\|Sx-Sy\| + \lambda_2\|Tx-Ty\| \qquad (x,y \in \mathcal{X}),$$

but T is not invertible. Consequently the endpoint $\lambda_2 = 1$ in Theorem 2.13 of [1] *is sharp.*

Proof. Let $\mathcal{X} = \mathcal{Y} = \ell_2$ and let $U : \ell_2 \to \ell_2$ be the unilateral shift

$$U(x_1, x_2, x_3, \ldots) = (0, x_1, x_2, x_3, \ldots).$$

Put $S = I_{\ell_2}$. Choose

$$\alpha \geq \frac{1 - \lambda_1}{\lambda_2 - 1}$$

and set $T = \alpha U$. Then T is a bounded linear, hence Lipschitz, map. It is not onto, since the first coordinate of every vector in its range is zero. Therefore T is not invertible as a map from ℓ_2 onto ℓ_2.

It remains only to check the perturbation inequality. For $h = x - y$,

$$\|Tx - Ty - (Sx - Sy)\| = \|\alpha Uh - h\|.$$

Since U is an isometry, the triangle inequality gives

$$\|\alpha Uh - h\| \leq \alpha\|Uh\| + \|h\| = (\alpha + 1)\|h\|.$$

The choice of α is exactly the condition

$$\alpha + 1 \leq \lambda_1 + \lambda_2\alpha.$$

Also $\|Tx - Ty\| = \alpha\|Uh\| = \alpha\|h\|$ and $\|Sx - Sy\| = \|h\|$. Hence

$$\|\alpha Uh - h\| \leq (\lambda_1 + \lambda_2\alpha)\|h\| = \lambda_1\|Sx - Sy\| + \lambda_2\|Tx - Ty\|.$$

This verifies all hypotheses of the would-be improved theorem with this $\lambda_2 > 1$, while the conclusion of invertibility fails. $\square$

Lemma 1. *In fact, for $\alpha > 0$, $\|\alpha U - I\| = \alpha + 1$.*

Proof. The upper bound is the triangle inequality used above. For the lower bound, let

$$h^{(n)} = \frac{1}{\sqrt{n}}(1, -1, 1, -1, \ldots, (-1)^{n-1}, 0, 0, \ldots).$$

Then $\|h^{(n)}\| = 1$ and a direct calculation shows that

$$\|(\alpha U - I)h^{(n)}\|^2 = \frac{1}{n}\left(1 + (n-1)(\alpha+1)^2 + \alpha^2\right).$$

Letting $n \to \infty$ gives $\|\alpha U - I\| \geq \alpha + 1$. $\square$

Verification notes

The argument is analytic and does not depend on numerical evidence. The crop script in `code/make_open_problem_crop.py` was used only to render the source-provenance image.

Cheap local indexes were searched for the arXiv id and the main keywords before promotion. A bounded web search on 2026-06-20 for the exact question, the paper title, and sharpness of the λ_2 endpoint found only the source arXiv page itself among relevant hits. The claim should be reviewed as a sharpness counterexample to the standard endpoint-extension reading of Question 2.14.

References

[1] K. Mahesh Krishna, *Improving Casazza-Kalton-Christensen-van Eijndhoven Perturbation with Applications*, arXiv:2109.02127, 2021.